

Precision Experimental Protocol for Testing Residual Force Signatures in Asymmetric Nonequilibrium Electromagnetic Systems

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Abstract

This work presents a precision experimental protocol for testing residual force signatures in asymmetric nonequilibrium electromagnetic systems. Building on the Emergent Vacuum Response Theory (EVRT) framework, the objective is not to claim anomalous force generation, but to define a controlled pathway for evaluating whether any residual force remains after known physical artifacts are eliminated.

The proposed protocol uses an asymmetric high-voltage electromagnetic test article mounted on a torsion balance or microbalance within a controlled vacuum environment. Experimental controls include polarity reversal, geometric reversal, symmetric dummy configurations, vacuum comparison, thermal monitoring, vibration isolation, electromagnetic shielding, and blind randomized power cycling.

A valid candidate signal must persist under vacuum, scale with electromagnetic energy or field asymmetry, survive dummy-load comparison, and remain reproducible across independent trials. Failure to satisfy these criteria results in a null classification, contributing to upper bounds on any effective response parameter.

This protocol is intended as a reproducible experimental foundation for testing speculative residual-force hypotheses while remaining grounded in established electrodynamics, conservation principles, and artifact rejection methodology.

1 Introduction

Asymmetric electromagnetic systems have historically motivated interest because apparent residual forces may arise under high-voltage, resonant, or nonequilibrium operating conditions. However, such systems are also especially vulnerable to conventional artifacts, including electrohydrodynamic effects, thermal drift, vibration, cable forces, electromagnetic coupling, and measurement bias.

Emergent Vacuum Response Theory (EVRT) provides a constrained phenomenological framework for investigating whether structured electromagnetic systems may exhibit small residual stress-energy effects beyond conventional expectations. The framework does not assert discovery of new physics. Instead, it defines experimentally testable conditions under which any proposed residual effect must be evaluated.

The present work defines a precision experimental protocol for testing residual force signatures in asymmetric nonequilibrium electromagnetic systems. The central goal is to determine whether any measurable signal persists after systematic elimination of known artifacts.

2 Experimental Objective

The experiment is designed to test the following question:

Can an asymmetric nonequilibrium electromagnetic system produce a reproducible residual force that survives vacuum operation, polarity reversal, geometric reversal, thermal stabilization, vibration isolation, electromagnetic shielding, and dummy-load comparison?

The expected baseline outcome is null. Any non-null result must satisfy strict discrimination criteria before being considered a candidate for further investigation.

3 Null Hypothesis

The null hypothesis assumes that all measured forces are explained by standard physical mechanisms:

$$\chi_{\text{eff}} = 0. \tag{1}$$

Under this condition, any observed force is attributed to known effects such as:

- electrohydrodynamic forces
- Lorentz forces on conductors
- thermal expansion or radiation imbalance
- mechanical vibration
- charge buildup or leakage currents
- Measurement system drift

A result is classified as null unless it survives all required control tests.

4 Test Article

The initial test article should be deliberately simple. A recommended configuration is an asymmetric capacitor assembly consisting of:

- one large electrode
- one smaller opposing electrode
- fixed dielectric or vacuum gap
- rigid nonconductive support structure
- shielded high-voltage feedthrough

The purpose of this geometry is not to maximize force, but to provide a clean asymmetric electromagnetic configuration suitable for artifact rejection.

5 Experimental Apparatus

A preferred experimental setup consists of:

- vacuum chamber
- torsion balance or high-sensitivity microbalance
- high-voltage power supply
- remote-controlled switching system
- thermal sensors near the test article
- vibration monitoring
- shielded electrical feedthroughs
- optical or capacitive displacement readout

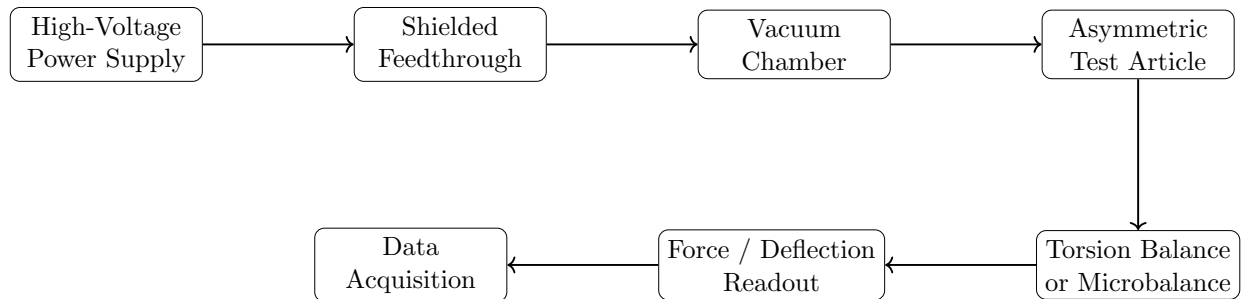


Figure 1: Conceptual experimental layout for residual-force testing in asymmetric nonequilibrium electromagnetic systems.

6 Operating Conditions

Initial operating conditions should be conservative:

- voltage range: 5–30 kV
- pressure range: atmospheric pressure to rough vacuum, with high vacuum preferred where available
- trial duration: 10–30 minutes per run
- repeated power cycling: off/on/off sequences
- polarity reversal after each run set
- device reversal by 180°

The force sensitivity target should be at least micronewton-scale for early trials, with nanonewton-scale sensitivity preferred for meaningful bounds.

7 Measurement Protocol

Each experimental run should follow the sequence:

1. Record baseline force or deflection with power off.
2. Apply high voltage under controlled ramp conditions.
3. Record force, temperature, vibration, current leakage, and pressure.
4. Switch power off and monitor relaxation behavior.
5. Repeat with reversed polarity.
6. Repeat with device rotated by 180° .
7. Repeat using a symmetric control device.
8. Repeat using a dummy electrical load.

A signal that appears only in the asymmetric device but not in dummy or symmetric controls is not sufficient by itself. It must also survive the full artifact rejection sequence.

8 Required Controls

Artifact Class	Expected Signature	Rejection Test
Thermal drift	Slow monotonic force change correlated with temperature	Thermal stabilization and multi-point temperature logging
Mechanical vibration	Periodic or phase-dependent force signal	Vibration isolation and accelerometer monitoring
Electromagnetic coupling	Force correlated with cable routing, grounding, or external fields	Shielding, grounding checks, and cable rerouting
Electrohydrodynamic effects	Signal present in air but reduced or absent in vacuum	Atmospheric vs. vacuum comparison
Cable forces	Apparent thrust changes with cable tension or placement	Mechanical cable isolation and dummy-load testing
Charge buildup	Delayed relaxation or hysteresis after power-off	Discharge control and leakage-current monitoring
Measurement system drift	Baseline offset unrelated to test article state	Blind runs and long-duration baseline tracking

Table 1: Summary of major artifact classes, expected signatures, and rejection tests for residual-force experiments.

Table 1 summarizes the minimum artifact-discrimination logic required before any residual-force signal can be considered a candidate effect.

Figure 2 illustrates the control-matrix logic used to systematically eliminate conventional artifact pathways. In practice, most observed signals are expected to be rejected at one or more intermediate stages.

8.1 Polarity Reversal

Polarity reversal tests whether the observed signal depends on electric-field orientation or instead arises from polarity-independent heating or mechanical effects.

8.2 Geometric Reversal

Rotating the device by 180° tests whether the signal tracks the device geometry. A valid geometry-linked signal should transform predictably under reversal.

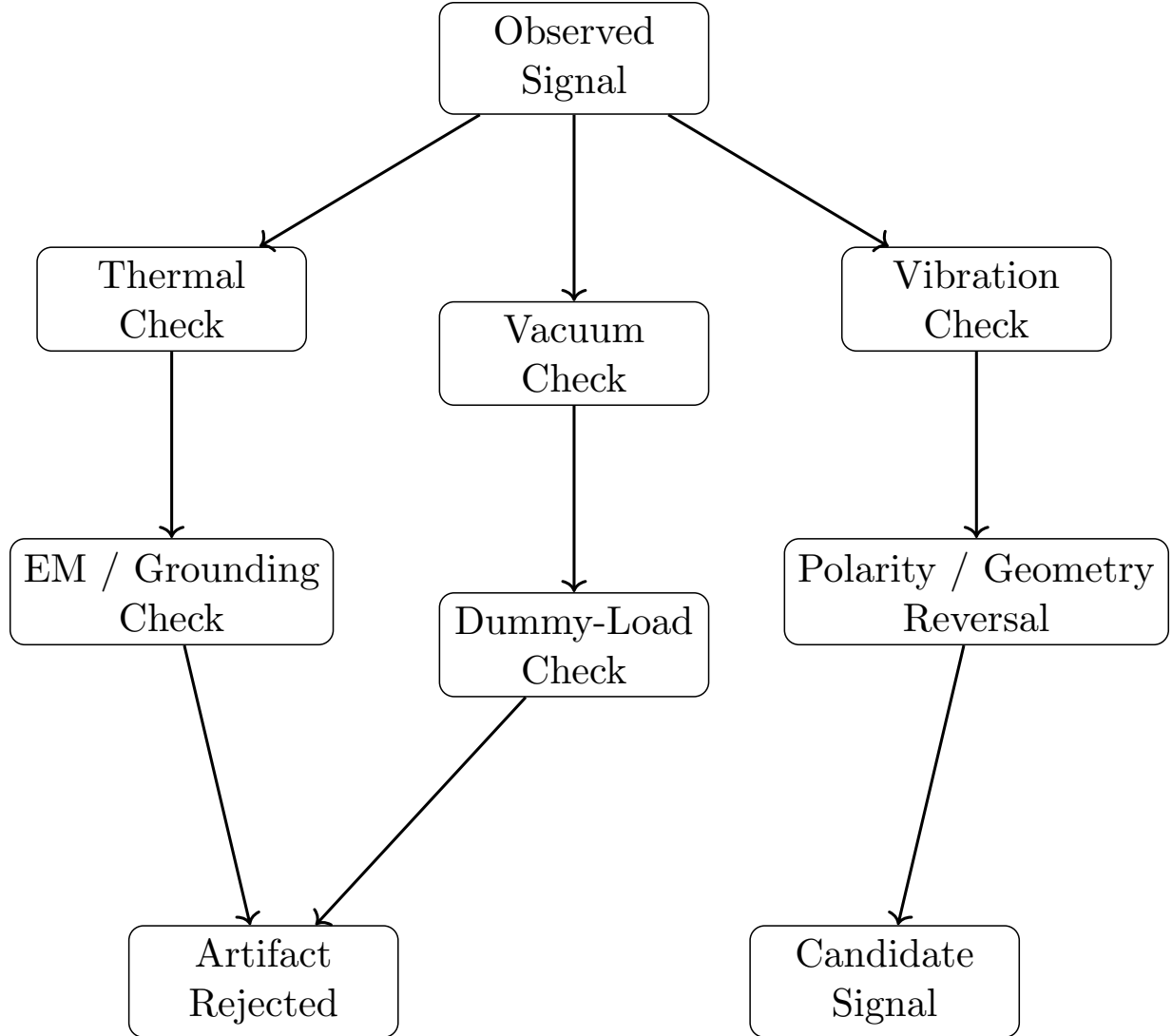


Figure 2: Control matrix for residual-force testing. A signal must survive all listed control stages to be considered a candidate residual-force effect. Failure at any stage favors a conventional artifact interpretation.

8.3 Symmetric Control

A symmetric capacitor or resonator should be tested under comparable electrical conditions. Any persistent force in the symmetric control argues against an EVRT-consistent interpretation.

8.4 Dummy Load

A dummy resistor or equivalent electrical load should reproduce current, heating, and cable conditions without producing the asymmetric field geometry. Signals appearing in the dummy case indicate instrumentation or environmental artifacts.

8.5 Vacuum Comparison

Atmospheric, rough vacuum, and high-vacuum operation should be compared. Signals that disappear in vacuum are likely associated with electrohydrodynamic effects.

8.6 Thermal Monitoring

Temperature changes must be recorded near electrodes, supports, cables, and the balance system. Correlation between force and thermal gradients should be treated as evidence for conventional artifact dominance.

8.7 Cable Isolation

High-voltage cables must be mechanically isolated from the balance. Cable motion, electrostatic attraction, or Lorentz forces can produce apparent thrust-like signals.

9 Signal Classification

A measured signal is classified as null if:

- it is statistically indistinguishable from baseline noise
- it appears in dummy-load controls
- it disappears under vacuum
- it correlates with thermal drift
- it changes with cable placement
- it fails polarity or geometric reversal tests

A signal may be classified as a candidate only if it satisfies all of the following:

- persists in vacuum
- exceeds statistical and systematic uncertainty
- scales with electromagnetic energy or field asymmetry
- survives dummy-load comparison
- transforms predictably under geometric reversal
- remains reproducible across repeated trials

10 Expected Scaling

If a non-null signal is present, it should exhibit qualitative scaling with stored electromagnetic energy. For a simple electric-field-dominated system, the electromagnetic energy density is approximately

$$u \sim \frac{1}{2} \epsilon_0 E^2. \quad (2)$$

A phenomenological residual force scale may be written as

$$F_{\text{res}} \sim \chi_{\text{eff}} \frac{uV}{L}, \quad (3)$$

where V is the active volume and L is a characteristic length scale.

This scaling is not presented as a prediction of confirmed new physics, but as an operational way to compare experimental sensitivity with effective parameter bounds.

11 Data Requirements

Each run should record:

- timestamp
- voltage
- current
- pressure
- force or displacement
- temperature at multiple points
- vibration amplitude
- polarity state
- device orientation
- control condition

A minimal dataset should be provided in machine-readable format, such as CSV, with clear metadata.

12 Analysis Method

The analysis should compare powered and unpowered intervals while accounting for drift and environmental variables. A conservative detection threshold should require:

$$|F_{\text{on}} - F_{\text{off}}| > 3\sigma_{\text{total}}, \quad (4)$$

where σ_{total} includes statistical noise and estimated systematic uncertainty. Any result below this threshold is treated as null.

13 Interpretation of Outcomes

13.1 Null Result

A null result constrains the effective response parameter χ_{eff} within the tested configuration. Such a result is scientifically meaningful because it reduces the allowed parameter space.

13.2 Candidate Signal

A candidate signal does not establish new physics. It identifies a configuration requiring further testing, independent replication, and expanded artifact analysis.

14 Discussion

The proposed protocol emphasizes falsification rather than confirmation. Most apparent residual-force signals in early-stage experiments are expected to be caused by conventional effects. The strength of the protocol lies in its requirement that any candidate signal survive a sequence of increasingly restrictive controls.

This protocol is intentionally conservative: the burden of proof is placed on the signal to survive successive artifact rejection stages rather than on the experimenter to interpret ambiguous results.

This approach aligns with the broader EVRT program by treating null results as useful constraints and non-null results as provisional candidates requiring independent verification.

15 Conclusion

This paper defines a precision experimental protocol for testing residual force signatures in asymmetric nonequilibrium electromagnetic systems. The protocol uses vacuum operation, reversal tests, dummy controls, thermal monitoring, vibration isolation, and electromagnetic shielding to distinguish candidate signals from conventional artifacts.

The objective is not to demonstrate anomalous force generation, but to establish a reproducible pathway for testing whether any residual effect remains after known physical mechanisms are eliminated. In this framework, both null and candidate outcomes contribute to the scientific evaluation of EVRT-like hypotheses.

Future work will focus on experimental implementation and reporting of measured constraints under this protocol, regardless of whether results are null or candidate signals.

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